TRISEP DS TS40

Food & Dairy NF Elements

TRISEP sanitary membrane elements for food & dairy applications have a long history of outstanding, stable field performance. TRISEP TS40 is a versatile nanofiltration membrane that offers high solute rejection of both salts and uncharged organic solutes while operating at lower pressures than reverse osmosis membranes. In many water purification applications, TS40 is considered a "softening" membrane, and these elements operate at a pressure of about 100 psi. TRISEP DS elements are built with FDA compliant materials, are compliant with European food regulations (EU 10/2011 and EC 1935/2004) and comply with USDA standards.

MEMBRANE CHARACTERISTICS

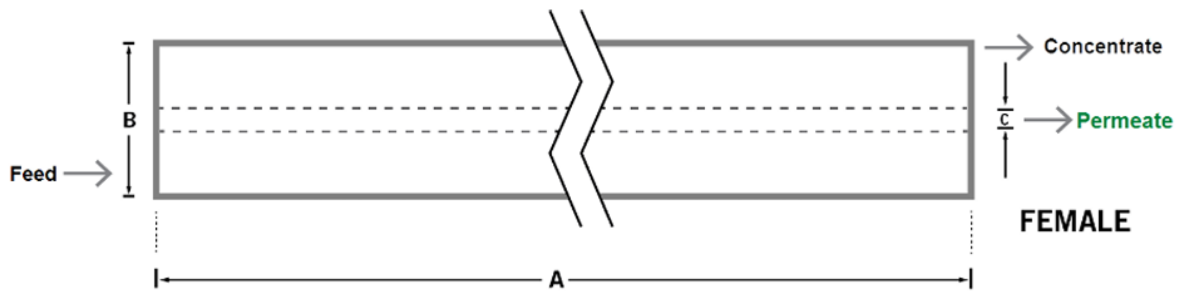
Membrane	TS40
Membrane Chemistry	Polypiperazine
Stabilized Salt Rejection (%)[a]	99.0
Minimum Salt Rejection (%)	98.5

a Typical NaCl rejection: 40%

DESIGN INFORMATION

Model[a]	Membrane Area m^2 (ft^2)	Feed Spacer Thickness (mil)[b]
TRISEP DS TS40 8038-65	21.8 (235.4)	17 (75)

a Test conditions: 2,000 ppm $MgSO_4$, 7.6 bar (110 psi), 25°C (77°F), 15% recovery, pH 8.0, 30 minutes operation. Flow rates will be no more than 15% below the values shown. Product specifications may change without notice as design revisions occur.
b All models on this sheet have a netting outer wrap and diamond shaped feed spacers.



PHYSICAL DIMENSIONS

Model	Element Weight kg (lb)[c]	Dim. A mm (inches)	Dim. B mm (inches)	Dim. C[d] mm (inches)	Permeate Tube
TRISEP DS TS80 8038-31	16 (36)	965 (38.0)	201 (7.9)	28.9 (1.125)	Female

c Shipping weight is dependent on packaging material and quantity shipped
d Dimension "C" is the Inner Diameter

OPERATING PARAMETERS

Maximum Operating Pressure	40 bar (580 psi)
Maximum Operating Temperature	50°C (122 °F)
Cleaning pH range[1]	1.0 -12.0
Cleaning Chlorine Tolerance[2]	< 0.1 ppm
Maximum Pressure Drop	1 bar (15 psi) per element; 4 bar (60 psi) per housing

1 Refer to temperature and pH limits in Membrane Cleaning Guide - Water Application Elements (TSG-C-001).
2 Pretreatment is recommended for the removal of free chlorine and other oxidizing agents to prevent damage to membranes. Oxidizing agents, such as free chlorine, in contact with polyamide membranes may result in shortened operating life or membrane failure. Such oxidation damage is excluded from warranty. Refer to Membrane Operating Guide - Recommendations for Water Purification (TSG-O-012).

IMPORTANT INFORMATION

Start-up: MANN+HUMMEL Water & Membrane Solutions recommends flushing elements for 30 minutes at low pressure and discarding permeate during the flush prior to operation. For a more detailed start-up procedure, please see Element Start-Up Guide System Start-Up (TSG-O-005)

Cleaning: TRISEP membrane elements must be cleaned periodically to ensure proper operation and to prevent membrane damage. Please see Membrane Cleaning Guide Food & Dairy: UF & MF Elements (TSG-C-004).

Storage: TRISEP membrane elements must be stored appropriately to ensure proper operation and to prevent membrane damage. Please see Element Storage Guides (TSG-O-009 & TSG-O-010).

Regulatory: All models on this sheet conform to USDA 3A sanitary standard 45-03, use FDA (CFR Title 21) compliant materials, comply with EU regulation (EC) No. 1935/2004 and No. 10/2011, and have Halal and Kosher certifications.

CUSTOMIZABLE SPECIALTY ELEMENTS

MANN+HUMMEL Water & Membrane Solutions offers a full range of membranes and element designs for challenging water and process applications. Technologies include low-fouling RO, submerged and pressurized UF, continuous high temperature, ultra-high pressure, unique sanitary designs and more. Contact us to customize a product that satisfies your specific requirement

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